

Correlation

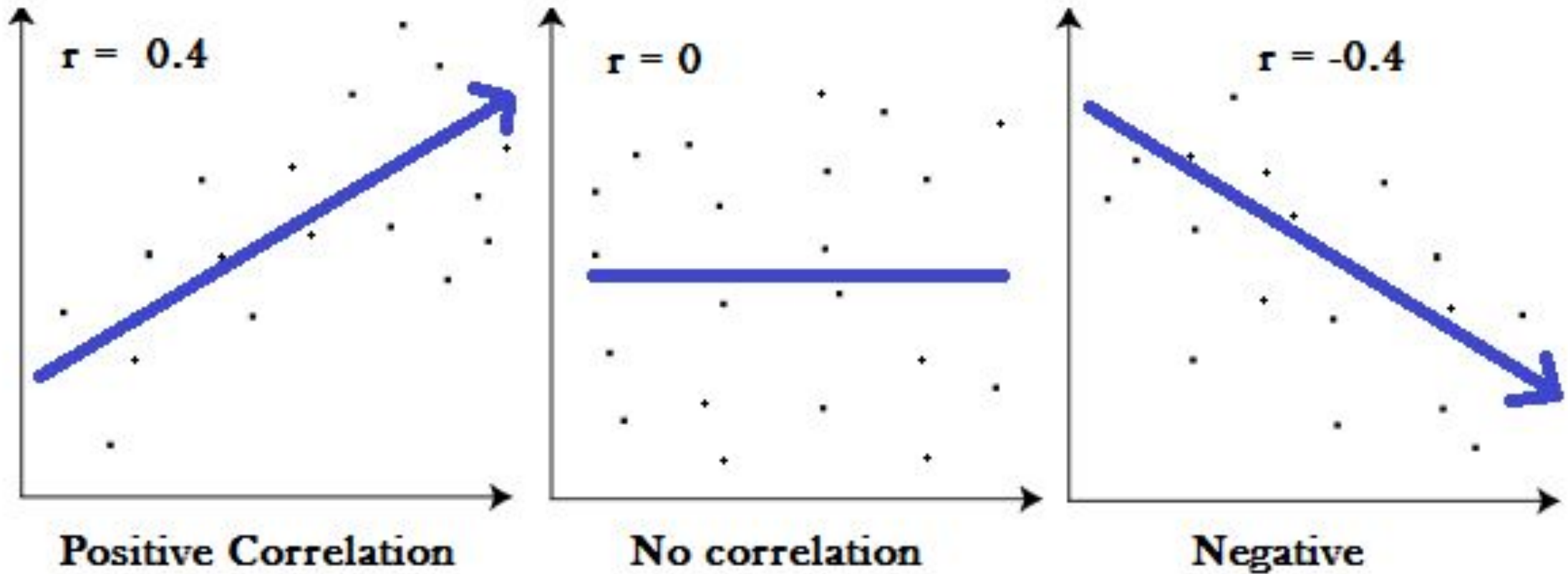
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Pearson's correlation

Pearson's correlation is a measure of the strength and direction of linear association that exists between two continuous variables (interval / ratio scale).

Correlation coefficient

- 1 indicates a strong positive relationship.
- -1 indicates a strong negative relationship.
- A result of zero indicates no relationship at all.



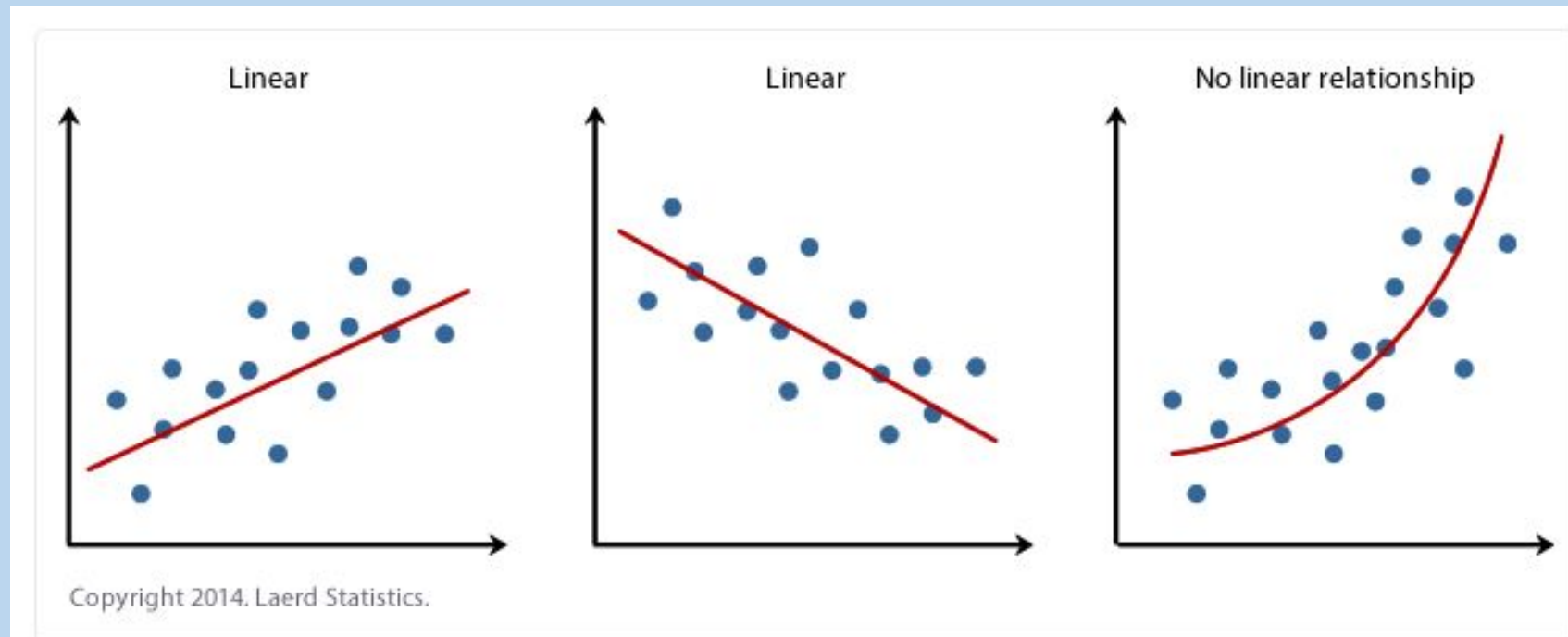
Graphs showing a correlation of -1, 0 and +1

- A correlation coefficient of 1 means that for every positive increase of 1 in one variable, there is a positive increase of 1 in the other.
- A correlation coefficient of -1 means that for every positive increase of 1 in one variable, there is a negative decrease of 1 in the other.
- Zero means that for every increase, there isn't a positive or negative increase. The two just aren't related.

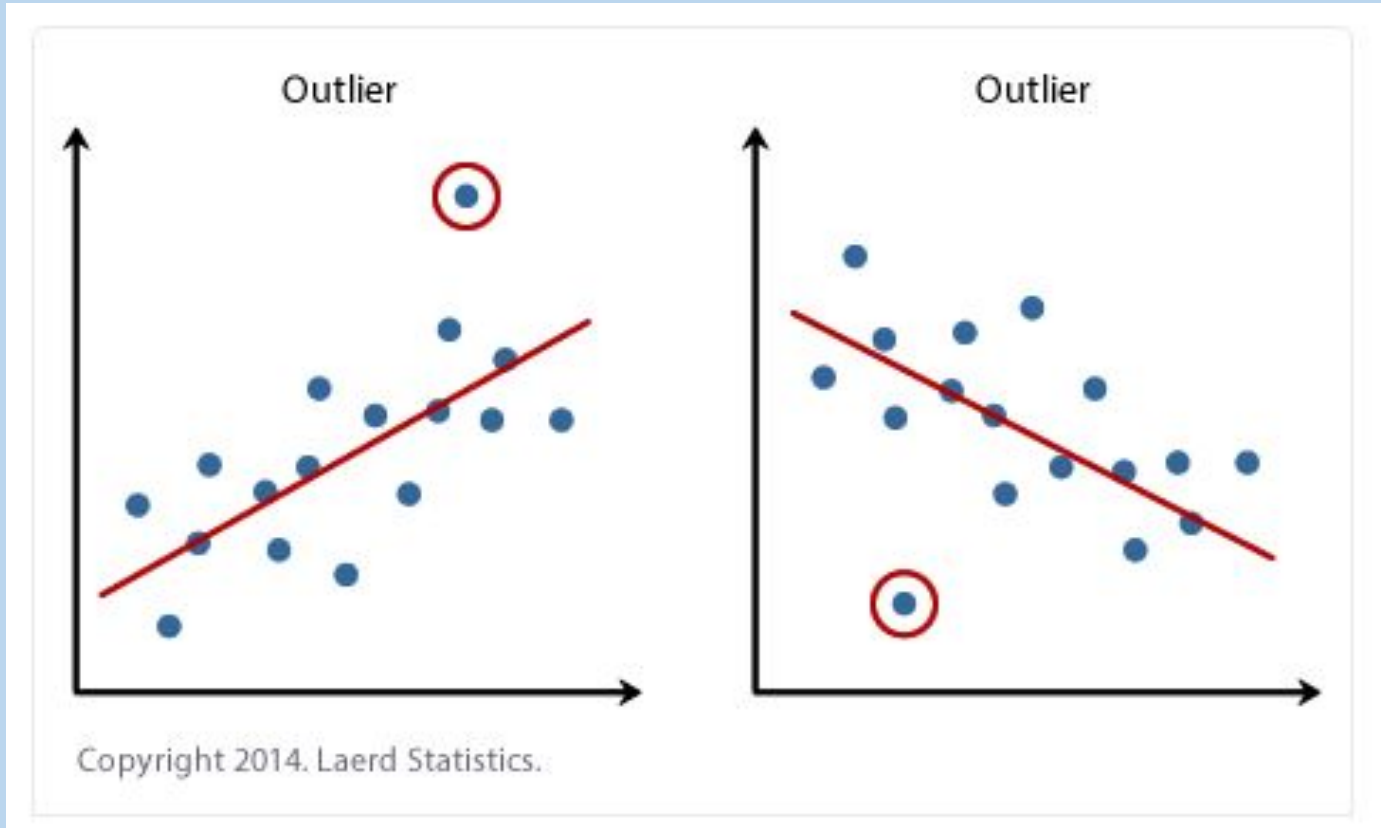
The **absolute value** of the correlation coefficient gives us the relationship strength. The larger the number, the stronger the relationship. For example, $|- .75| = .75$, which has a stronger relationship than .65.

4 assumptions must be satisfied for doing a correlation analysis

1. Your two variables (both dependent and independent variables) should be measured at the **interval** or **ratio level** (i.e., they are **continuous**).
2. there is a **linear relationship** between the two variables;



3. there are no significant outliers



4. Your both variables are **normally distributed**.

Test for normality of the study variables

If Normally distributed -----> Pearson correlation

Significance value of Shapiro Wilk test – is more than 0.05 for both the dependent and independent variables – means that the variables are normally distributed

[DataSet1] - IBM SPSS Statistics Data Editor

Transform Analyze Direct Marketing Graphs Utilities Add-ons W

age

27

63

36

26

24

29

24

27

25

22

30

45

25

36

23

29

37

30

36

36

26

maxAerobicCapacity

55.79

35.00

43.83

43.48

30.38

40.17

36.01

44.22

38.76

33.09

44.81

31.94

34.48

47.23

45.06

55.12

45.58

37.52

Reports

Descriptive Statistics

Tables

Compare Means

General Linear Model

Generalized Linear Models

Mixed Models

Correlate

Regression

Loglinear

Neural Networks

Classify

Dimension Reduction

Scale

Nonparametric Tests

Forecasting

Survival

Multiple Response

Missing Value Analysis...

Multiple Imputation

Complex Samples

Simulation...

Quality Control

ROC Curve...

Bivariate...

Partial...

Distances...

Bivariate Correlations

Variables:

serial number [slnr]

Age [age]

Heart Rate [heart_ra...]

Gender [gender]

MaxAerobicCapacity ...

Weight [weight]

Options...

Style...

Bootstrap...

Correlation Coefficients

☒ Pearson ☐ Kendall's tau-b ☐ Spearman

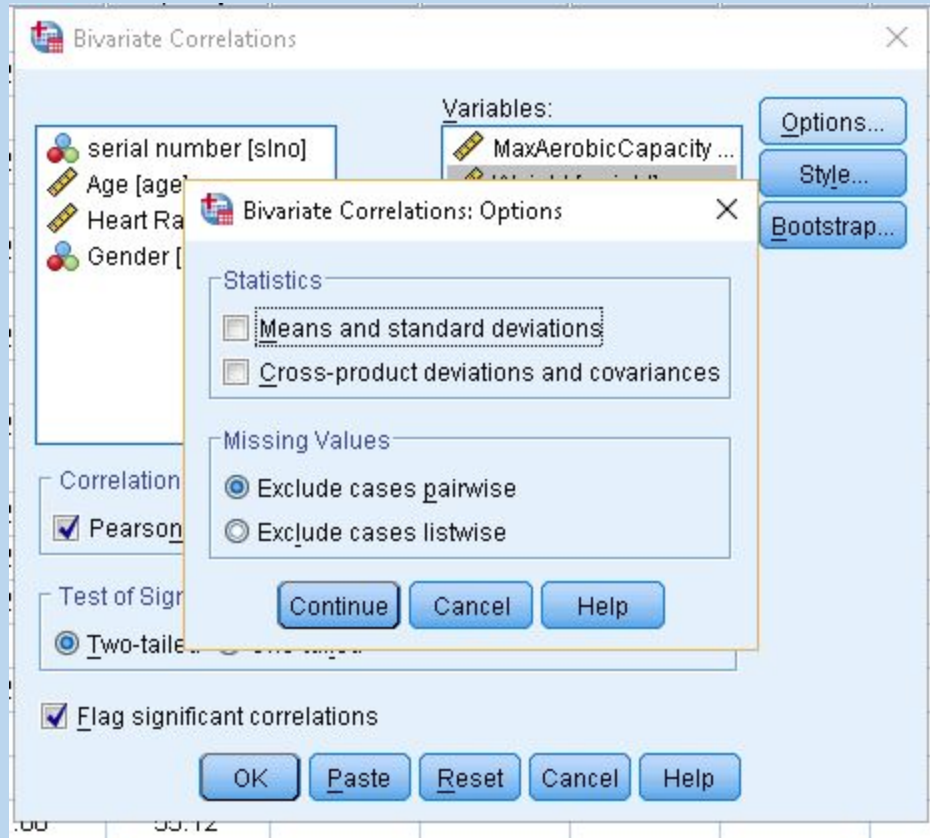
Test of Significance

☒ Two-tailed ☐ One-tailed

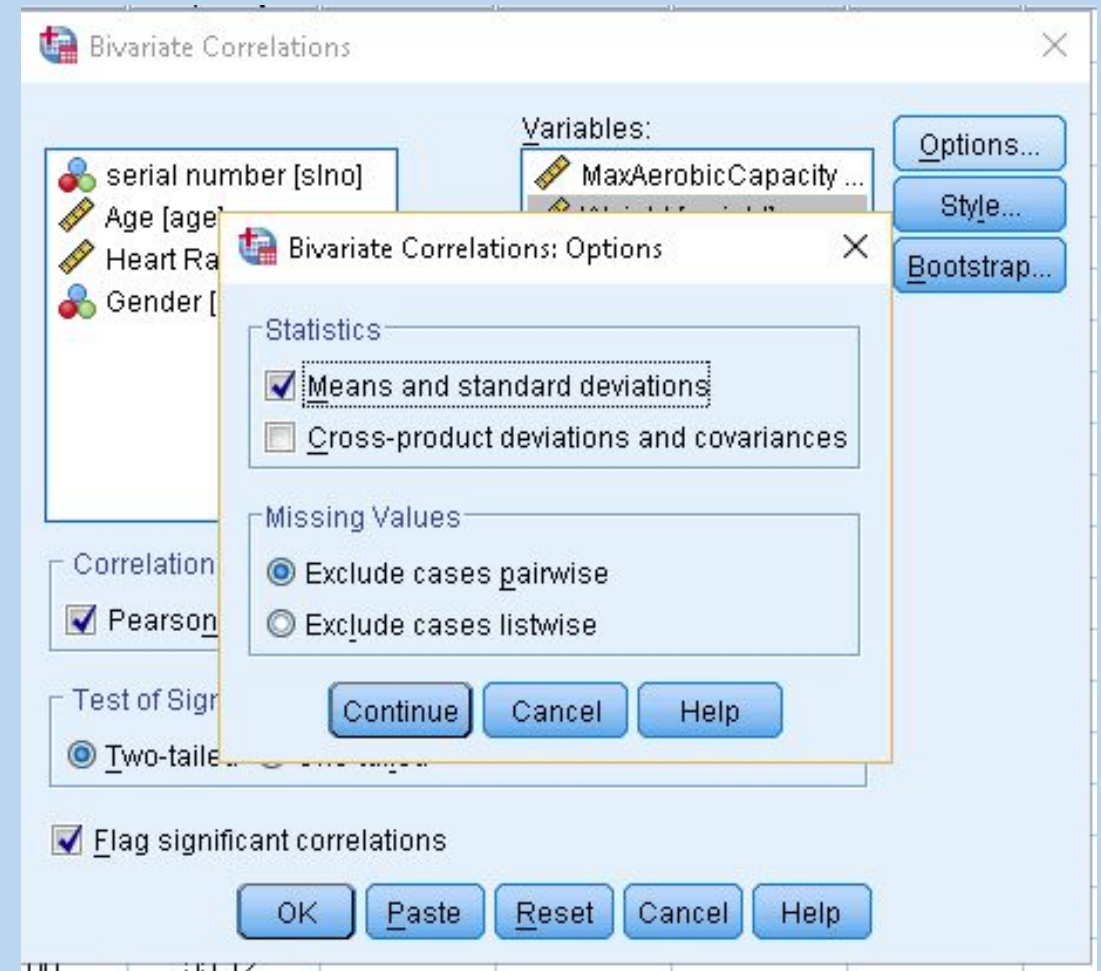
☒ Flag significant correlations

OK Paste Reset Cancel Help

Select variables



Click options



Check means and standard deviations

Click OK -----> The results appears in output viewer

Descriptive Statistics

	Mean	Std. Deviation	N
MaxAerobicCapacity	43.6298	8.57131	100
Weight	79.6603	15.08984	100

Correlations

		MaxAerobicC apacity	Weight
MaxAerobicCapacity	Pearson Correlation	1	-.307**
	Sig. (2-tailed)		.002
	N	100	100
Weight	Pearson Correlation	-.307**	1
	Sig. (2-tailed)	.002	
	N	100	100

** . Correlation is significant at the 0.01 level (2-tailed).

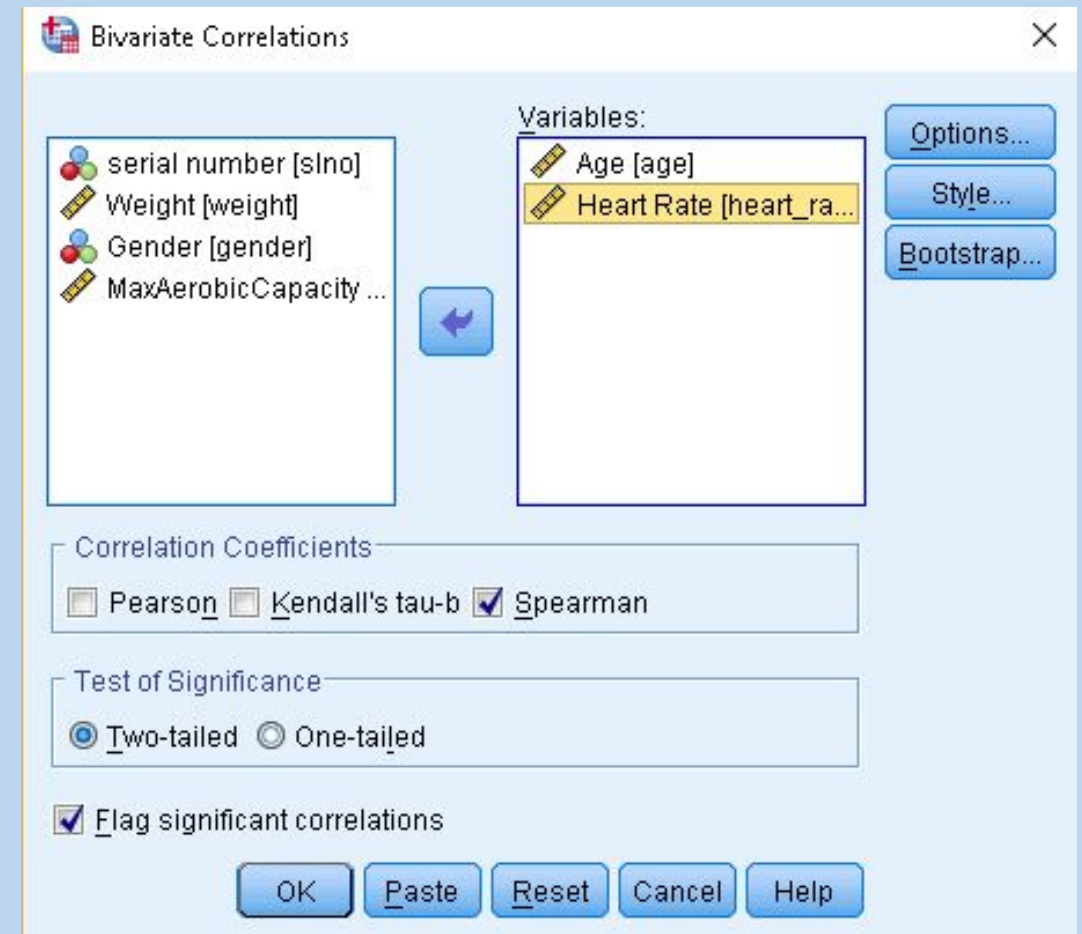
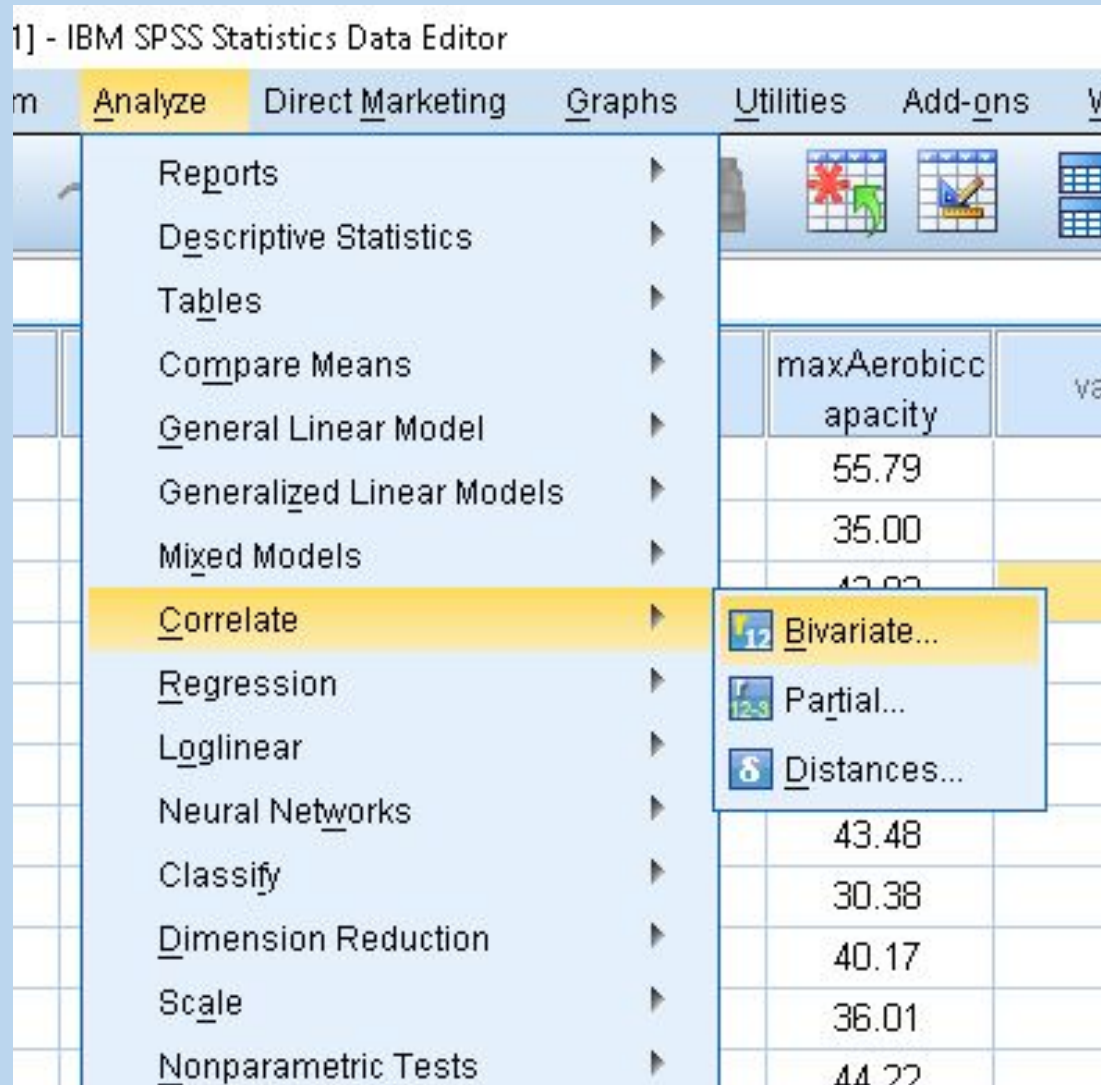
Spearman's correlation

Spearman's correlation is a measure of the strength and direction of association that exists between two continuous variables (interval / ratio scale).

Test for normality of the study variables

If any of the 2 variables

not Normally distributed -----> Spearman's correlation



Select variables
Click Spearman correlation

Click OK

Nonparametric Correlations

Correlations

			Age	Weight
Spearman's rho	Age	Correlation Coefficient	1.000	.057
		Sig. (2-tailed)	.	.573
		N	100	100
	Weight	Correlation Coefficient	.057	1.000
		Sig. (2-tailed)	.573	.
		N	100	100